

Cristhiano Duarte, Ph.D.

Advisory Board Member - Fundação Maurício Grabois

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Personal Information

I am a researcher whose work focuses on: **Quantum Information Theory, Foundations of Probability and Foundations of Physics**. I work as a Research Fellow at the Federal Univesity of Bahia and hold a position at the Maurício Grabois Foundation's Curator Board. I was a visiting researcher, fully funded by the FQXi at the International Institute of Physics (Natal - Brazil). I was also a research fellow with the School for Physics and Astronomy at the University of Leeds. Before settling in the UK, I was a visiting researcher at the prestigious Wigner Centre for Physics (Budapest - Hungary). At Chapman University I held a postdoctoral position mixing teaching and research in a program called Grand Challenges Initiative. Before that I was a postdoc at the International Institute of Physics (Natal - Brazil). I was also a postdoc at two other institutions: right after my PhD I got a position at the Federal University of Minas Gerais (Brazil), following that I got a (brief summer postdoctoral) position at the Brazilian Institute for Pure and Applied Mathematics (Rio de Janeiro - Brazil).

Education

- 2013 – 2016 ■ **Ph.D. Mathematics, Federal University of Minas Gerais, Brazil.**
Title: The Parts and the Whole: how the concept of tensor product affects quantum theory.
- 2011 – 2013 ■ **M.Sc. Mathematics, Federal University of Juiz de Fora, Brazil.**
Title: On Entanglement Catalysis in Quantum Mechanics.
- 2006 – 2011 ■ **B.Sc. Physics, Federal University of Juiz de Fora, Brazil.**
Project title: Can All Entangled States be Useful?

Grants

- 2025 ■ **CNPq - Conhecimento Brasil - Linha 1**
Grant title: Quantum Theory as a Theory of Probabilistic Assignment *Principal Invesitgators:* Cristhiano Duarte *Total Amount:* 1,3 million BRL
- **CNPq - Conhecimento Brasil - Linha 2**
Grant title: Quantum Theory as a Theory of Probabilistic Assignment *Principal Invesitgators:* Marcelo Terra Cunha e Cristhiano Duarte *Total Amount:* 1,3 million BRL
- 2021 ■ **FQXi - Outreach Grant**
Grant title: Impededance Collective - Minorities in Physics *Principal Invesitgators:* Nadja K. Bernardes, Gabriela B. Lemos. Facilitated via FXQi by **Cristhiano Duarte** *Total Amount:* 7k US dollars

Grants (continued)

- 2019 **FQXi - Intelligence in the Physical World**
Grant title: Reasoning in the quantum world. Principal Investigator: Cristhiano Duarte
co-PI: Matt Leifer Total Amount: 135k US dollars
- 2018 **Open Arms Brazil - International Congress of Mathematicians 2018**
Grant title: Open Arms Brazil Travel Grant Awardee: Cristhiano Duarte
Declined in Protest: Before the realisation of the ICM-2018 the conference organisers published an open letter supporting another case of police brutality in Rio de Janeiro.

Teaching Experience

- 2013 **Temporary Teacher - Federal University of Minas Gerais.**
Undergraduate level. Discipline: Analytic Geometry and Linear Algebra - Engineering School
- Temporary Teacher - Federal University of Minas Gerais.**
Undergraduate level. Discipline: Analytic Geometry and Linear Algebra - Economic and Business Faculty/Physics Faculty
- 2014 **Voluntary Teacher - Federal University of Minas Gerais.**
Undergraduate level. Discipline: How to construct the real numbers.
Special course: planned to those students that had been awarded with medals in the Brazilian maths Olympiad.
- 2018 **Grand Challenges Initiative Instructor - Chapman University**
Undergraduate level. Discipline: FFC100-03 and FFC100-04.
Grand Challenges Initiative Program: Entry point of a project oriented series of courses emphasising critical thinking, critical use of source of information, conceptual synthesis, scientific communication and teamwork.
- 2019 **Grand Challenges Initiative Instructor - Chapman University**
Undergraduate level. Discipline: SCI200-01,SCI200-04, SCI200-06 and SCI200-12.
Grand Challenges Initiative Program: Third semester of a project oriented series of courses emphasising critical thinking, critical use of source of information, conceptual synthesis, scientific communication and teamwork.
- 2020 **Grand Challenges Initiative Instructor - Chapman University**
Undergraduate level. Discipline: SCI250-02.
Grand Challenges Initiative Program: Last semester of a project oriented series of courses emphasising critical thinking, critical use of source of information, conceptual synthesis, scientific communication and teamwork.
- Grand Challenges Initiative Instructor - Chapman University**
Undergraduate level: Discipline: SCI150-01 and SCI150-02.
Grand Challenges Initiative Program: Second semester of a project oriented series of courses emphasising critical thinking, critical use of source of information, conceptual synthesis, scientific communication and teamwork.

Supervision and Mentoring

PhD Supervision

- 2020  **Carlos Humberto Vieira - Campinas State Univeristy**
Title: On the impossibility of broadcasting and cloning resourceful objects
Start date: 2019 End date: 2023
Supervisors: Cristhiano Duarte and Marcelo Terra Cunha
- 2021  **Adrian Solyomos - Wigner Center**
Title: TBD
Start date: 2019 End date: 2024 (Tentative)
Supervisor: Zoltan Zimboras
Co-Supervisor (Informal): Cristhiano Duarte

Master's Thesis Supervision

-  **Zhen Cai - Imperial College of London**
Title: Quantum Darwinism for Quantum Channels
Start date: 2020 End date: 2022 (Concluded)
Supervisor: Cristhiano Duarte
- 2022  **Silvio Junior - Sao Paulo State University**
Title: Efficient Quantum Tomography via Machine Learning
Start date: 2019 End date: 2024 (Tentative)
Supervisor: Barbara Amaral
Co-Supervisor: Cristhiano Duarte

Research Publications

Journal Articles

- 1 Nery, R., Bernardes, N. K., Cavalcanti, D., Chaves, R. & **Duarte, C.** (2025). Efficient and operational quantifier of nondivisibility in terms of channel discrimination. *Phys. Rev. A*, 111, 022206. <https://doi.org/10.1103/PhysRevA.111.022206>
- 2 Azevedo, T. M. D., **Duarte, C.** & Bernardes, N. K. (2024). Distillation of quantum non-markovianity. *Physics Letters A*, 512, 129582. <https://doi.org/10.1016/j.physleta.2024.129582>
- 3 Valle, V. G., Brugger, L. L., Rizzuti, B. F. & **Duarte, C.** (2024). Towards establishing a connection between two-level quantum systems and physical spaces. *Brazilian Journal of Physics*, 54(4), 93. <https://doi.org/10.1007/s13538-024-01462-6>
- 4 Grossi, R., Brugger, L. L., Rizzuti, B. & **Duarte, C.** (2023). One hundred years later: Stern-gerlach experiment and dimension witnesses. *Revista Brasileira de Ensino de Física*, 45, e20220227. <https://doi.org/10.1590/1806-9126-rbef-2022-0227>
- 5 Poderini, D., Rodari, G., Moreno, G., Polino, E., Nery, R., Suprano, A., **Duarte, C.**, Sciarino, F. & Chaves, R. (2023). Device-independent witness for the nonobjectivity of quantum dynamics. *Phys. Rev. A*, 108, 032201. <https://doi.org/10.1103/PhysRevA.108.032201>
- 6 Acácio, T. M. & **Duarte, C.** (2022). Analysis of neural network predictions for entanglement self-catalysis. *Brazilian Journal of Physics*, 52(6), 189. <https://doi.org/10.1007/s13538-022-01191-8>

- 7 **Duarte, C.**, Catani, L. & Drumond, R. C. (2022). Relating compatibility and divisibility of quantum channels. *International Journal of Theoretical Physics*, 61(7), 189. <https://doi.org/10.1007/s10773-022-05165-z>
- 8 Moreno, G., Nery, R., **Duarte, C.** & Chaves, R. (2022). Events in quantum mechanics are maximally non-absolute. *Quantum*, 6, 785. <https://doi.org/10.22331/q-2022-08-24-785>
- 9 Baldijão, R. D., Wagner, R., **Duarte, C.**, Amaral, B. & Cunha, M. T. (2021). Emergence of noncontextuality under quantum darwinism. *PRX Quantum*, 2, 030351. <https://doi.org/10.1103/PRXQuantum.2.030351>
- 10 Vieira, C. H. S., **Duarte, C.**, Drumond, R. C. & Terra Cunha, M. (2021). Bell non-locality in many-body quantum systems with exponential decay of correlations. *Brazilian Journal of Physics*. <https://doi.org/10.1007/s13538-021-00998-1>
- 11 Åberg, J., Nery, R., **Duarte, C.** & Chaves, R. (2020). Semidefinite tests for quantum network topologies. *Phys. Rev. Lett.*, 125, 110505. <https://doi.org/10.1103/PhysRevLett.125.110505>
- 12 **Duarte, C.** (2020). Compatibility between agents as a tool for coarse-grained descriptions of quantum systems. *Journal of Physics A: Mathematical and Theoretical*. <https://doi.org/10.1088/1751-8121/aba574>
- 13 Leifer, M. & **Duarte, C.** (2020). Noncontextuality inequalities from antidistinguishability. *Phys. Rev. A*, 101, 062113. <https://doi.org/10.1103/PhysRevA.101.062113>
- 14 Rizzuti, B. F., Gaio, L. M. & **Duarte, C.** (2020). Operational approach to the topological structure of the physical space. *Foundations of Science*. <https://doi.org/10.1007/s10699-020-09650-8>
- 15 Amaral, B. & **Duarte, C.** (2019). Characterizing and quantifying extended contextuality. *Phys. Rev. A*, 100, 062103. <https://doi.org/10.1103/PhysRevA.100.062103>
- 16 Rai, A., **Duarte, C.**, Brito, S. & Chaves, R. (2019). Geometry of the quantum set on no-signaling faces. *Phys. Rev. A*, 99, 032106. <https://doi.org/10.1103/PhysRevA.99.032106>
- 17 Amaral, B., **Duarte, C.** & Oliveira, R. I. (2018). Necessary conditions for extended non-contextuality in general sets of random variables. *Journal of Mathematical Physics*, 59(7), <https://doi.org/10.1063/1.5024885>, 072202. <https://doi.org/10.1063/1.5024885>
- 18 **Duarte, C.** & Amaral, B. (2018). Resource theory of contextuality for arbitrary prepare-and-measure experiments. *Journal of Mathematical Physics*, 59(6), <https://doi.org/10.1063/1.5018582>, 062202. <https://doi.org/10.1063/1.5018582>
- 19 **Duarte, C.**, Brito, S., Amaral, B. & Chaves, R. (2018). Concentration phenomena in the geometry of bell correlations. *Phys. Rev. A*, 98, 062114. <https://doi.org/10.1103/PhysRevA.98.062114>
- 20 **Duarte, C.**, Drumond, R. C. & Oliveira, R. I. (2018). Small violations of bell inequalities for multipartite pure random states. *Journal of Mathematical Physics*, 59(5), 052202. <https://doi.org/10.1063/1.5026690>
- 21 **Duarte, C.**, Carvalho, G. D., Bernardes, N. K. & de Melo, F. (2017). Emerging dynamics arising from coarse-grained quantum systems. *Phys. Rev. A*, 96, 032113. <https://doi.org/10.1103/PhysRevA.96.032113>
- 22 Drumond, R. C., **Duarte, C.**, Terra Cunha, M. & Nemes, M. C. (2016). Universality of finite-time disentanglement. *Phys. Rev. A*, 93, 022313. <https://doi.org/10.1103/PhysRevA.93.022313>

- 23 **Duarte, C.**, Drumond, R. C. & Cunha, M. T. (2016). Self-catalytic conversion of pure quantum states. *Journal of Physics A: Mathematical and Theoretical*, 49(14), 145303. <http://stacks.iop.org/1751-8121/49/i=14/a=145303>
- 24 Bragança, H., Mascarenhas, E., Luiz, G. I., **Duarte, C.**, Pereira, R. G., Santos, M. F. & Aguiar, M. C. O. (2014). Nonuniversality of entanglement convertibility. *Phys. Rev. B*, 89, 235132. <https://doi.org/10.1103/PhysRevB.89.235132>
- 25 Rabelo, R., **Duarte, C.**, López-Tarrida, A. J., Cunha, M. T. & Cabello, A. (2014). Multi-graph approach to quantum non-locality. *Journal of Physics A: Mathematical and Theoretical*, 47(42), 424021. <http://stacks.iop.org/1751-8121/47/i=42/a=424021>

Submitted Articles

- 1 Leifer, M. & **Duarte, C.** (2022). *Generalising aumann's agreement theorem*. arXiv. <https://doi.org/10.48550/ARXIV.2202.02156>
- 2 **Duarte, C.**, Amaral, B., Cunha, M. T. & Leifer, M. (2020). *Investigating coarse-grainings and emergent quantum dynamics with four mathematical perspectives*. arXiv. <https://doi.org/10.48550/ARXIV.2011.10349>

Miscellaneous Experience

Awards and Achievements

- 2018 ■ **Travel Grant**, Open Arms Brazil-Travel Grant – Grant awarded for participation in the International Congress of Mathematicians 2018 (declined for political reasons).
- 2017 ■ **Summer Postdoctoral Fellowship**, Institute for Pure and Applied Mathematics (IMPA) – Term spent as a Researcher under the supervision of Dr. Roberto Imbuzeiro at IMPA.

Academic Visits and Collaborations (partially or fully funded)

- 2023 ■ **Short Visit**: Federal University of Pernambuco (Visiting Professor) – Collaboration with Dr. Nadja Kolb Bernardes
- 2022 ■ **Short Visit**: Wigner Research Institute for Physics (Wigner Center) – Collaboration with Zoltan Zimboras, Carlos Humberto Vieira, and Adrian Solymos.
- **Short Visit**: Institute for Quantum Studies (Chapman University) – Collaboration with Alexander Kurz.
- 2020 ■ **Short Visit**: The Institute for Photonic Sciences (ICFO) – Collaboration with Dani Cavalcanti and Antonio Acin.
- 2019 ■ **Short Visit**: Perimeter Institute for Theoretical Physics (PI) – Collaboration with Robert Spekkens and Matt Leifer.
- 2018 ■ **Short Visit**: Institute for Pure and Applied Mathematics (IMPA) – Collaboration with Roberto Imbuzeiro.
- **Short Visit**: Brazilian Center for Research in Physics (CBPF) – Collaboration with Fernando de Melo.

Miscellaneous Experience (continued)

- 2017 **Visiting Specialist:** Brazilian Center for Research in Physics (CBPF) – Term spent as Visiting Specialist in the Quantum Information Group at CBPF.
- Short Visit:** Max-Planck Institute for Mathematics in the Sciences (MIS) – Collaboration with Tobias Fritz - Special Seminar: Sudden Death of Entanglement
- 2016 **Visiting Specialist:** Brazilian Center for Research in Physics (CBPF) – Term spent as Visiting Specialist in the Quantum Information Group at CBPF.

Referee Work

- 2021 **European Journal of Physics**
- 2020 **Quantum - The Open Journal for Quantum Science**
- 2017 **Journal of Physics A: Mathematical and Theoretical**

Editorial Work

- 2022 **Entropy**, Special Issue “Quantum Correlations, Contextuality, and Quantum Nonlocality” – Co-editor of the special issue.
- Princípios: Teoria, Política e Cultura**, Associated Scientific Editor.

Events

Talks, Poster Presentations and Participations at Events and Conferences

- 2024 **X Workshop da Pós-Graduação em Física . Title (talk):** *Generalising Aumann’s Agreement Theorem*. 2020. (Workshop)
- 2022 **II Workshop of Quantum Information and Foundations of Quantum Mechanics at UNICAMP (Encontrinho). Title (talk):** *Generalising Aumann’s Agreement Theorem*. 2020. (Workshop)
- 2020 **I Institute for Quantum Studies Workshop. Title (talk):** *Quantum Agency*. 2020. (Workshop)
- 2019 **VII Quantum Information School and Workshop - Paraty. Title (poster):** *Small Violations of Bell Inequality*. 2019. (School and Workshop)
- Quantum Physics and Logic (QPL-2019) - Chapman University - USA. Title (talk):** *Concentration phenomena in the geometry of Bell correlations*. 2019. (Conference)
- SYCO 4: fourth symposium in compositional structures - Orange - USA. 2019 (Symposium)**

Events (continued)

- PIMan Workshop: First Perimeter Institute and Chapman Workshop in Quantum Foundations - Orange - USA. **Title (talk):** *Small Violations of Bell-Inequalities*. 2019. (Workshop)
- APS March Meeting - Boston - USA. **Title (talk):** *Small Violations of Bell-Inequalities*. 2019. (Conference)
- Wigner's Friends in Boston - Boston - USA. 2019. (Workshop)
- 2018 ■ Q-Turn changing paradigms in quantum science - Florianopolis. **Title (talk):** *Small Violations Bell Inequalities*. 2018. (Workshop)
- X Summer Workshop of Mathematics at Federal University of Juiz de Fora. **Title (talk):** *Small Violations of General Bell Inequalities*. 2018. (Workshop)
- 2017 ■ Workshop on Quantum Correlations, Contextuality and All That... Again and Again Natal. 2017. (Workshop)
- VI Quantum Information School and Workshop - Paraty. **Title (poster):** *On the complete understanding of self-catalysis*. 2017. (School and Workshop)
- Quantumness: A celebration of Luiz Davidovich's research career on his 70th birthday. 2017. (Special Meeting).
- 2016 ■ VI Workshop de Informação e Óptica Quântica. **Title (talk):** *Emergent Coarse-Grained Dynamics*. 2016. (Meeting).
- 2015 ■ 30o Colóquio Brasileiro de Matemática. **Title (talk):** *Coloured-Graph Approach to Quantum Non-Locality*. 2015. (Congress).
- Quantum Information at Benasque. **Title (presentation):** *Coloured Graphs Approach to Quantum Non-Locality*. 2015. (Meeting).
- V Quantum Information School and Workshop. **Title (poster):** *Colored Graph Approach to Quantum Non-Locality*. 2015. (School and Workshop)
- Workshop on Quantum Correlations, Contextuality and All That... Again. 2015. (Workshop).
- 2013 ■ IV Quantum Information School and Workshop - Paraty. **Title (poster):** *Catalysis in Quantum Systems*. 2013. (School and Workshop)
- 2012 ■ I Workshop da Pós-Graduação em Física at Federal University of Juiz de Fora. **Title (talk):** *Catalysis in Quantum Systems*. 2012. (Workshop).
- 2011 ■ Jornadas em Geometria Algébrica-JOGA. 2011. (Congress).

Events (continued)

- 2010  V Simpósio Nacional/ Jornadas de Iniciação Científica. **Title (poster):** *Entanglement: an introduction*. 2010. (National Meeting).
-  WECIQ-2010: Workshop and School of Quantum Information. Petrópolis-Brazil. 2010. (Workshop).
- 2009  XXX Encontro Nacional de Física de Partículas e Campos. **Title (poster):** *String-Localized Quantum fields*. 2009. (Meeting).
- 2008  Workshop Quantum Vacuum-Quantum vacuum properties in laboratories and in the Universe. 2008. (Congress).

Organisation of Events

- 2020  Quantum Retreat in Natal - A Break from the Madness. 2022. (Workshop - Organiser)
-  II QTurn - Changing Paradigms in Quantum Science. 2020. (Conference - Scientific Committee and Referee)
- 2018  I QTurn - Changing Paradigms in Quantum Science . 2018. (Conference - Scientific Committee and Referee)
- 2016  VI Workshop de Informação e Óptica Quântica (VI Workshop of Quantum Information and Quantum Optics). 2016. (Meeting - Organiser)

Complementary Education

- 2022  **Ethics - An Introduction** - *Oxford University*
Duration: 10 weeks (100 hours worthy of work)
Online Platform: Oxford Continuing Education
-  **Political Philosophy** - *Oxford University*
Duration: 10 weeks (100 hours worthy of work)
Online Platform: Oxford Continuing Education
-  **Fundamentos - Focus Group on Colonialism and Authoritarianism** - *Instructor: Sabrina Fernandes, PhD*
Duration: 24 weeks
Online Platform: Hotmart
- 2021  **Scepticism - An Introduction** - *MOOC led by the University of California Irvine*
Duration: 8 weeks
Online Platform: Coursera
-  **Classical Sociological Theory** - *MOOC led by the University of Amsterdam*
Duration: 8 weeks
Online Platform: Coursera

Complementary Education (continued)

- 2020  **Introduction to Philosophy** - MOOC led by the University of Edinburgh
Duration: 5 weeks
Online Platform: Coursera
- 2012  **Riemannian Geometry** - Intensive Summer Course
Duration: 8 weeks
In-person Lectures: Federal University of Minas Gerais
- 2011  **Topology** - Intensive Summer Course
Duration: 8 weeks
In-person Lectures: Federal University of Juiz de Fora
- 2010  **Topics of Advanced Linear Algebra** - Intensive Summer Course
Duration: 8 weeks
In-person Lectures: Federal University of Juiz de Fora

References

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